MAJOR PROJECT

EduCloud – Trainer–Student Secure File Portal

Linux + AWS End-to-End Project

Student Details

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Course: AWS & Linux Administration

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ABSTRACT

EduCloud is a cloud-based educational platform designed to provide a secure, scalable, and highly available environment for the distribution of study materials and management of assignment submissions. The system is developed using Amazon Web Services (AWS) and demonstrates the practical implementation of modern cloud computing concepts in an educational environment. The architecture incorporates services such as Amazon VPC, EC2, Application Load Balancer (ALB), Auto Scaling, Amazon S3, Amazon CloudFront, IAM, AWS Lambda, and Amazon RDS to support reliable application delivery, secure data management, and automated operations. Role-based access controls are implemented to ensure that students, trainers, and support personnel can access only the resources relevant to their responsibilities. Automation is used for tasks such as file management, backups, notifications, and cloud-based workflows. The project emphasizes high availability, fault tolerance, security, scalability, performance optimization, and cost-effective resource utilization, demonstrating how AWS services can be integrated to build a practical and production-oriented cloud infrastructure.

INTRODUCTION

The rapid adoption of cloud computing in the education sector has created opportunities to improve the way learning resources, assignments, and academic data are stored, accessed, and managed. Traditional educational systems often rely on centralized servers and manual processes, which can lead to limited scalability, availability issues, inefficient resource management, and increased administrative effort. A cloud-based approach can overcome these limitations by providing on-demand resources, centralized storage, automated operations, and secure access from different locations.

EduCloud is developed to address these challenges by integrating multiple AWS services into a unified cloud infrastructure. The platform provides centralized storage for study materials and assignments using Amazon S3, while IAM enables controlled and role-based access to cloud resources. Amazon EC2, Application Load Balancer, and Auto Scaling are used to maintain application availability and dynamically handle changes in workload. Amazon CloudFront improves content delivery by distributing frequently accessed resources through edge locations, reducing latency for users. AWS Lambda supports event-driven automation and reduces the need for continuously running compute resources, while Amazon RDS provides managed database services for storing application-related information.

The project goes beyond basic application development and focuses on the design, deployment, security, automation, monitoring, and management of cloud infrastructure. It demonstrates how different AWS services can work together to create a reliable and scalable educational platform while following important cloud principles such as least-privilege access, fault tolerance, automation, resource optimization, and cost efficiency.

OBJECTIVES

The primary objectives of the EduCloud project are:

- Design a secure and scalable cloud architecture using appropriate AWS services and networking components.
- Implement role-based access control for students, trainers, and support personnel using AWS IAM.
- Provide centralized and reliable storage for study materials, assignments, and backup data using Amazon S3.
- Enable secure file management and assignment submission while controlling access according to user roles.
- Ensure high availability and fault tolerance through Application Load Balancer, Auto Scaling, and multi-AZ deployment.
- Improve content delivery performance using Amazon CloudFront and distributed content delivery.
- Automate cloud operations and workflows using AWS Lambda and scheduled automation mechanisms.

KEY COMPONENTS

The EduCloud platform is built using multiple AWS services, where each component performs a specific role in providing secure, scalable, highly available, and reliable cloud infrastructure.

- **Amazon CloudFront:** Distributes frontend content through AWS edge locations, reducing latency and improving the loading speed of web pages and static resources.
- **Application Load Balancer (ALB):** Acts as the entry point for application traffic and distributes incoming requests across multiple EC2 instances to improve availability and prevent overloading of individual servers.
- **Auto Scaling Group (ASG):** Automatically adjusts the number of EC2 instances according to application workload and configured scaling policies, helping maintain performance while optimizing resource usage.
- **Amazon EC2:** Provides the compute environment for hosting and running the EduCloud application and web services.
- **Amazon S3:** Provides durable and scalable object storage for study materials, assignment submissions, backups, and other educational resources.
- **AWS Lambda:** Executes event-driven and scheduled automation tasks without requiring dedicated servers. It can be used for backend processing, notifications, and other cloud workflows.

- **Amazon RDS:** Provides a managed relational database for storing structured application information and metadata while reducing database administration overhead.
- **Amazon VPC:** Provides an isolated virtual networking environment in which AWS resources can be organized into public and private network segments with controlled communication.

The architecture follows a layered approach, separating internet-facing components from backend resources. Public-facing resources such as the load balancer are placed in public subnets, while application servers and database resources can be isolated within private subnets. This separation helps reduce the attack surface and provides greater control over network traffic.

NETWORK FLOW

The general traffic flow in the EduCloud architecture can be represented as:

User → CloudFront → ALB → EC2 Instances → RDS

For resources requiring outbound internet access:

Private EC2 → NAT Gateway → Internet Gateway → Internet

The Internet Gateway is associated with the public subnet routing, while the NAT Gateway provides controlled outbound connectivity for resources located in private subnets.

SYSTEM OVERVIEW

1. Frontend EduCloud Portal
    a. Simple static portal hosted on Apache /var/www/html/index.html on Private EC2.
    b. Trainers: Upload Materials button Uploads lecture slides/PPT to /materials Auto sync to S3.
    c. Students: Download Materials button Downloads via CloudFront Signed URL. Submit Assignment .tar.gz via FTP.
    d. Support: View Logs button Monitors /logs for validation errors, access logs.
2. Linux Backend – Core Engine

a. Users & Groups:

- Trainers: trainer1, trainer2 (Group: trainers) - RW on /materials
- Students: student1, student2, student3 (Group: students) - RO on /materials, RW on /assignments

- Support: support1 (Group: support) - RO on /logs
- Password expiry 30 days via chage -M 30

b. File System & Security:

- File Permissions: ACLs enforce: /materials Trainers RW Students RO, /assignments Students RW Trainers RO, /logs Support RO.
- Storage: /training_data 5GB LVM volume Auto expands to 10GB at 85% full. /sensitive_data 2GB LUKS encrypted for exam_solutions.pdf.
- File Sharing: NFS shares /materials (Trainers RW, Students RO). FTP with TLS enabled jails students to /assignments only. Your main requirement /assignments is S3 mount via s3fs, so FTP upload = instant S3 storage.

c. Automation:

- trainer1: Cron 0 23 _ _ _ backup /materials daily 11 PM to S3
- support1: Cron 0 18 _ _ 0 upload logs Sundays 6 PM to S3
- trainer2: Cron 30 0 1 _ * rotate assignments monthly 1st 12:30 AM3.

d. AWS Cloud – Scalable Infrastructure

Network: VPC EduCloudVPC 10.0.0.0/16 - Public Subnet 10.0.1.0/24 (ALB + IGW) + Private Subnet 10.0.2.0/24 (EC2 + RDS + EBS) + NAT Gateway for private internet.

Flow:

- User request -> Internet -> CloudFront CDN (caches study materials)
- ALB (Application Load Balancer) distributes traffic
- ASG (2-5 EC2 Amazon Linux 2 Portal Servers) auto scales if CPU >60% for 5 mins
- Private EC2 accesses RDS MySQL educloud_db (Tables: students, trainers, uploads) + EBS 20GB + S3

VPC Structure

EduCloud uses an Amazon Virtual Private Cloud (VPC) to create an isolated and controlled networking environment for its cloud resources.

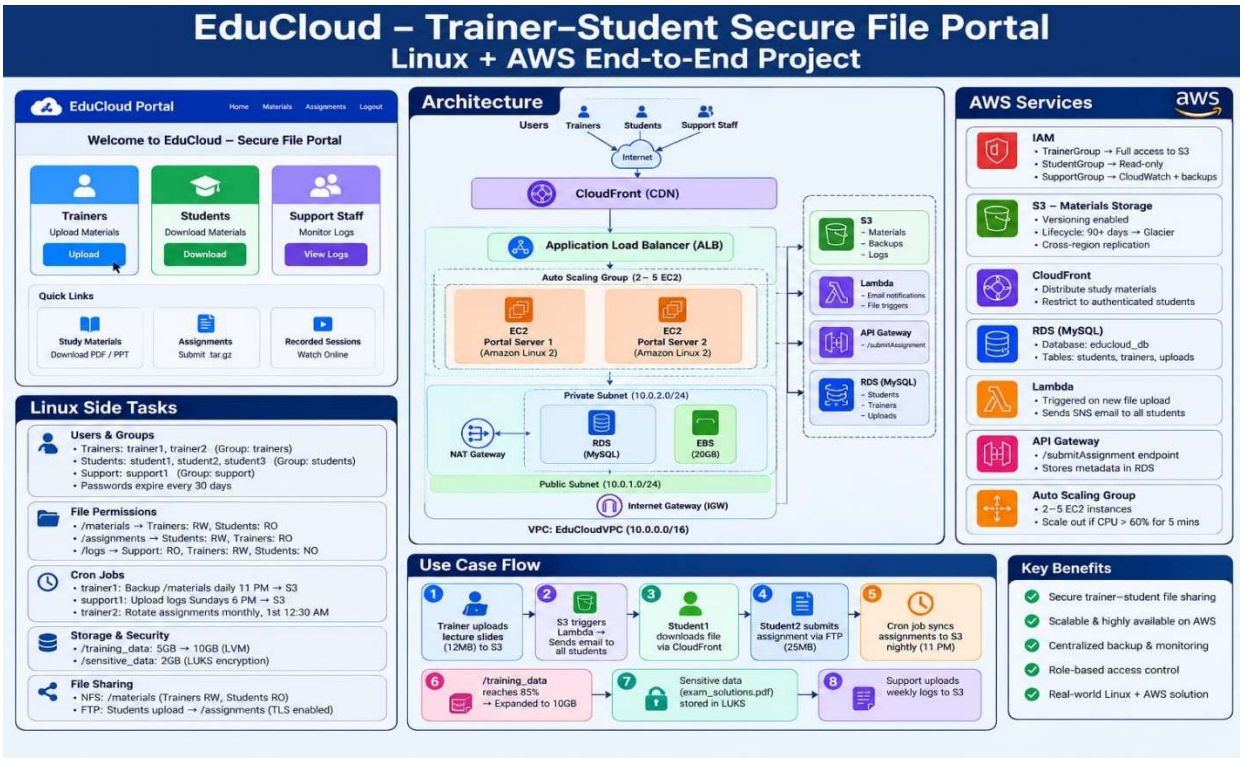
- **VPC CIDR Block:** 10.0.0.0/16
- **Public Subnets:** Used for internet-facing resources such as the Application Load Balancer.

- **Private Subnets:** Used for backend resources such as application servers and database services.
- **Internet Gateway (IGW):** Provides internet connectivity to resources in public subnets.
- **NAT Gateway:** Allows resources in private subnets to initiate outbound internet connections when required, without exposing those resources directly to inbound internet traffic.
- **Route Tables:** Define how traffic is routed between subnets, the Internet Gateway, NAT Gateway, and other network destinations.
- **Security Groups:** Act as virtual firewalls that control inbound and outbound traffic for individual AWS resources.

AWS Services:

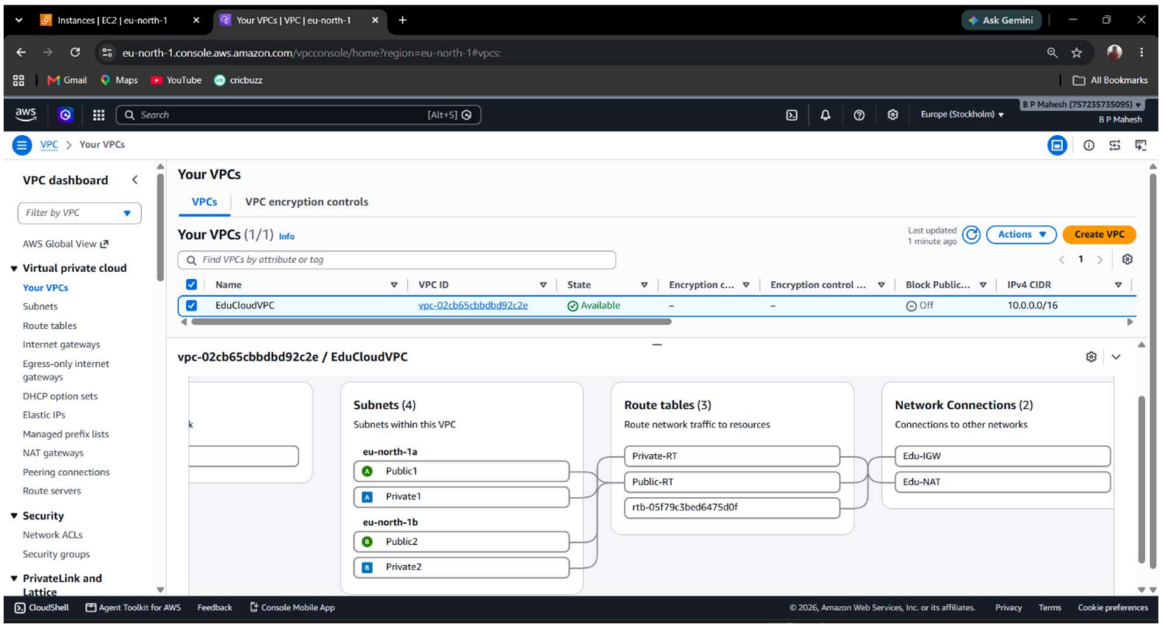
- IAM: TrainerGroup full S3, StudentGroup read-only, SupportGroup CloudWatch+backups
- S3: Materials storage with versioning, lifecycle 90 days -> Glacier, cross-region replication
- Lambda: Triggered on new S3 file upload -> Sends SNS email to all students
- API Gateway: /submitAssignment endpoint stores metadata in RDS
- CloudFront + EBS + RDS: Global fast delivery, persistent storage, central metadata DB

SYSTEM ARCHITECTURE

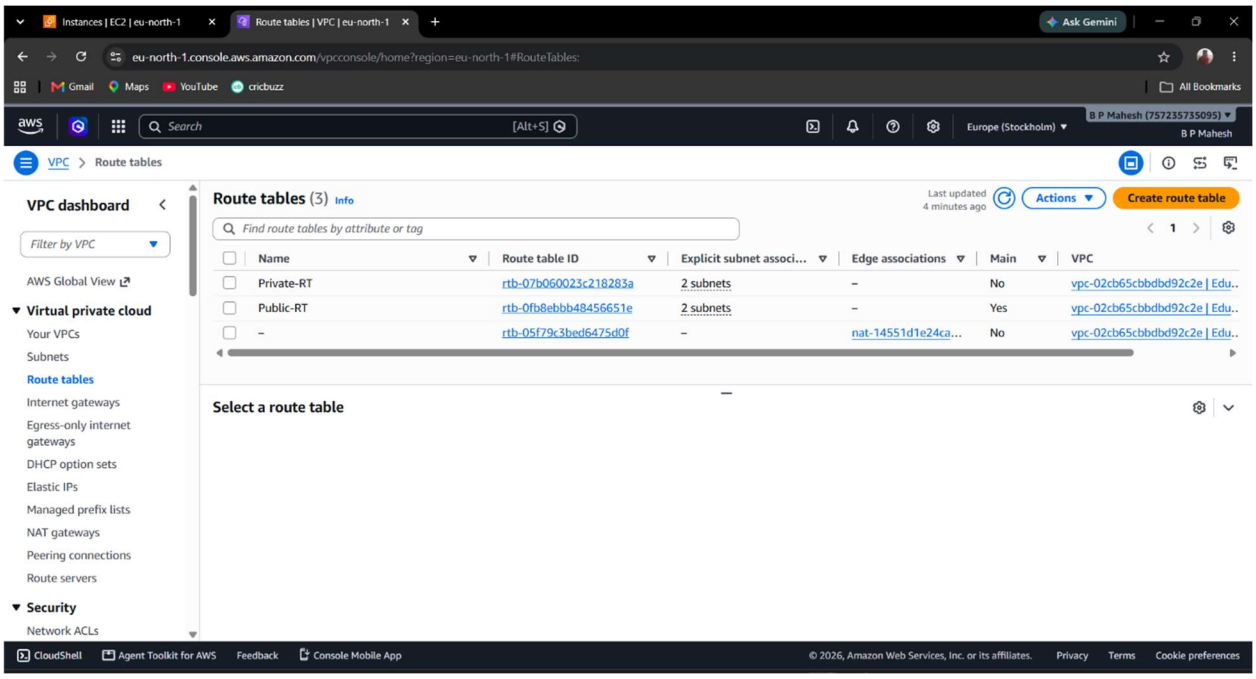


SCREENSHOTS / EVIDENCE

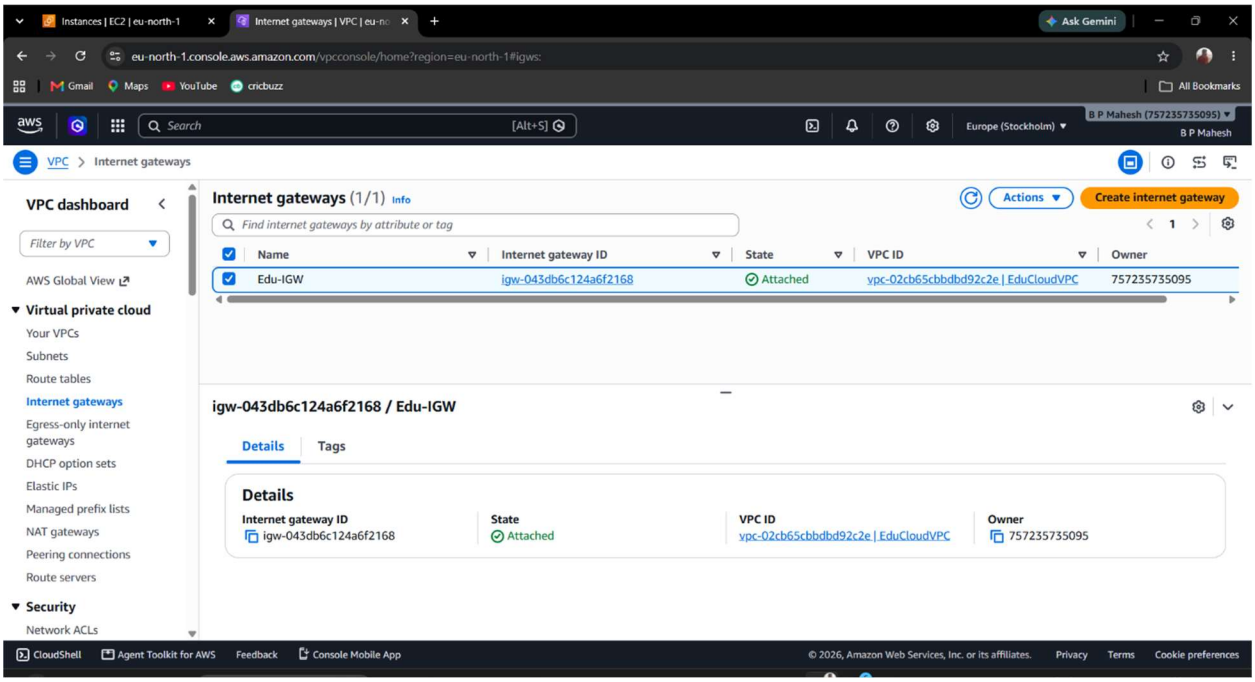
VPC And Subnet Configuration



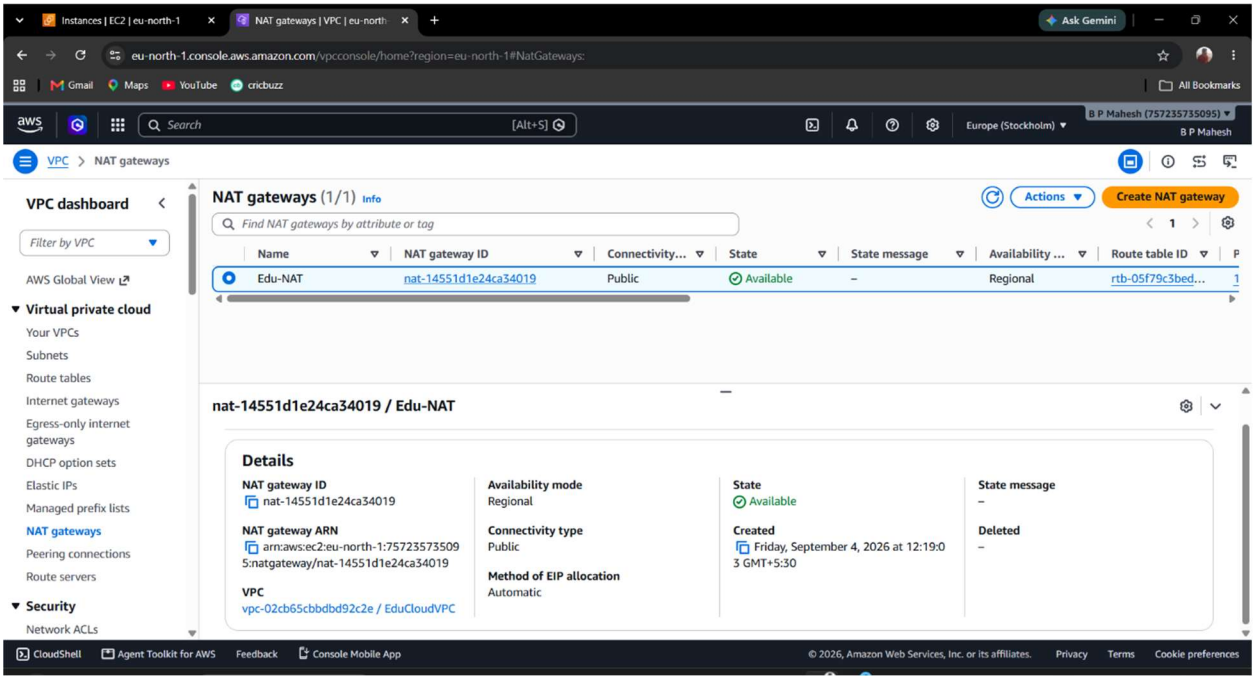
Route Table



Internet Gateway

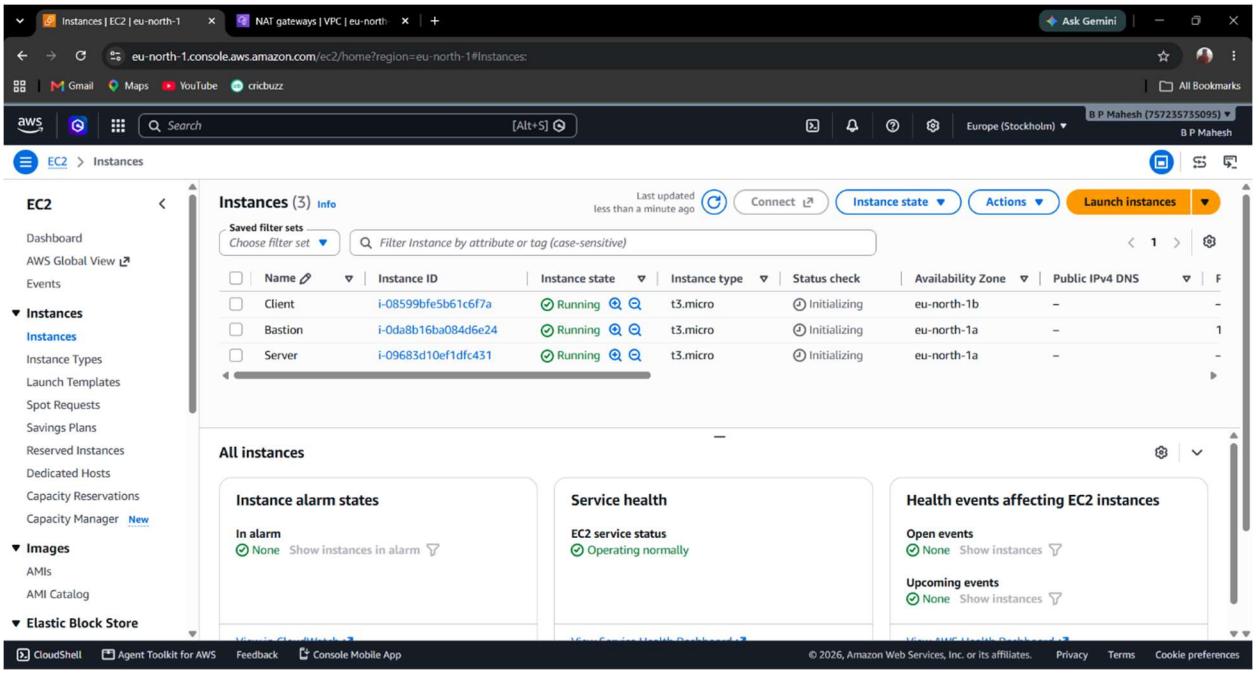


NAT Gateway

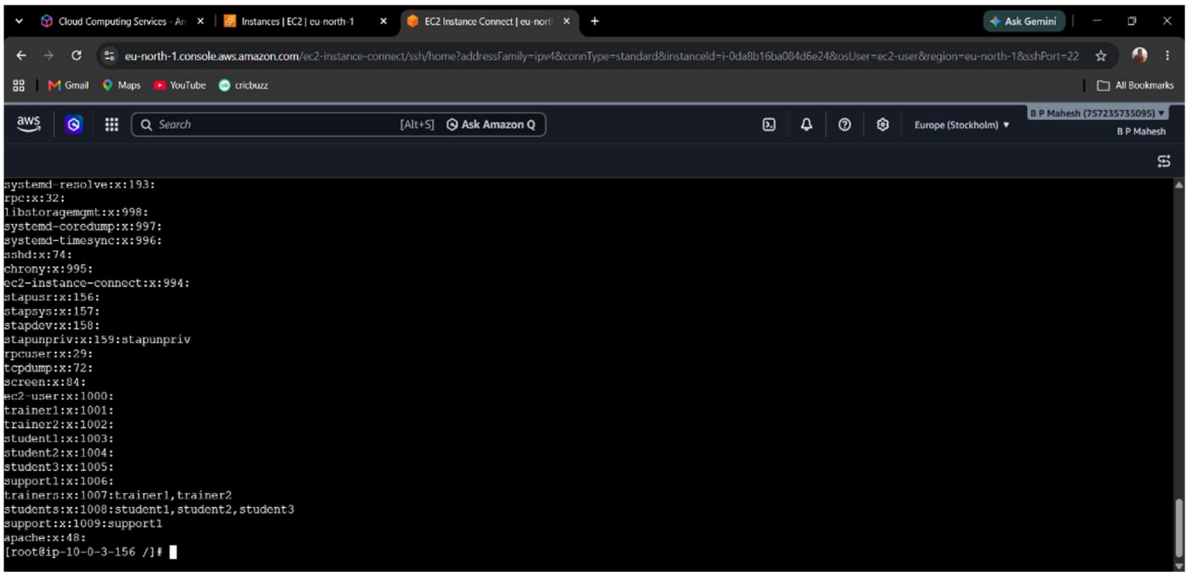


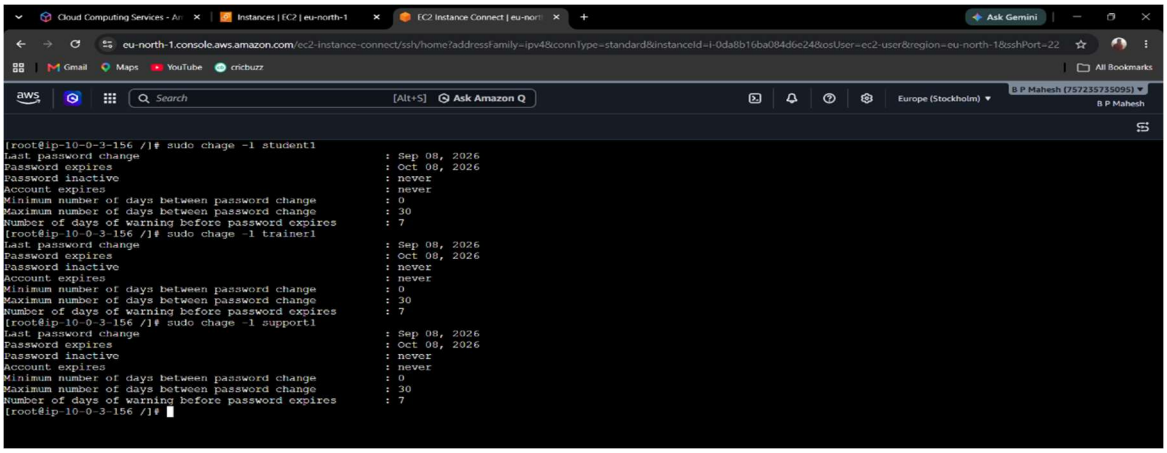
Linux Administration:

EC2-Instance



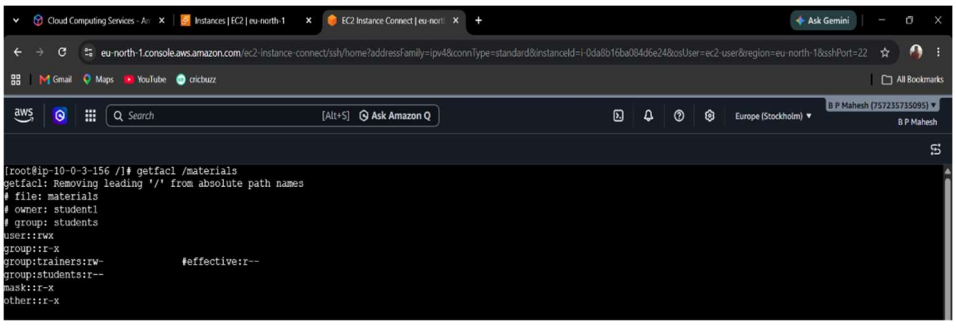
User and Group Creation





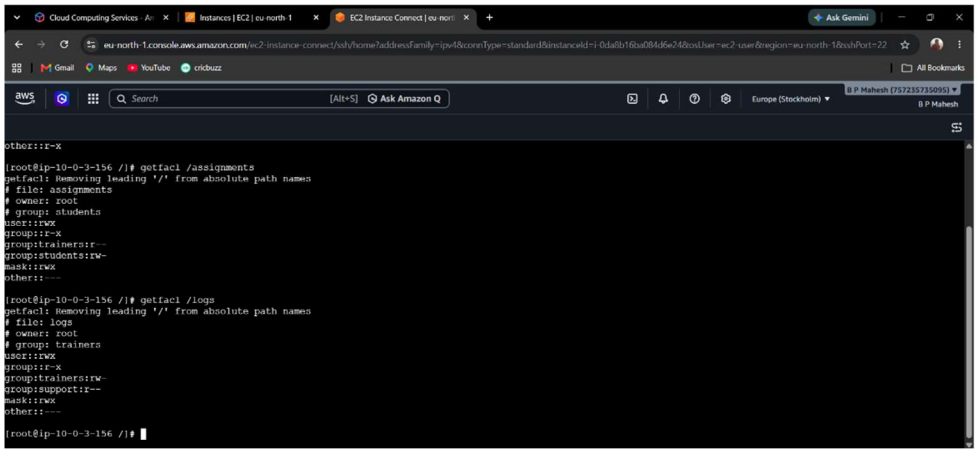
File Permissions & ACLs

- /materials → Trainers = read/write, Students = read-only.



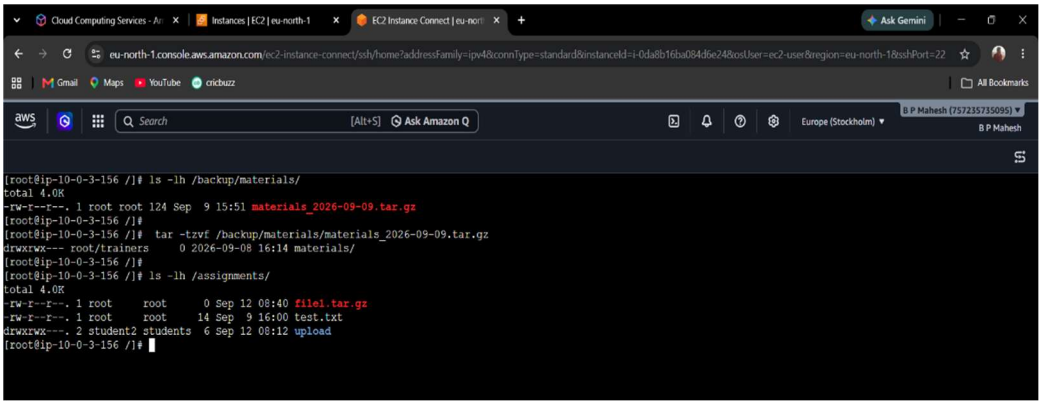
- /assignments → Students = read/write, Trainers = read-only.

- /logs → Support = read-only, Trainers = read/write, Students = no access.



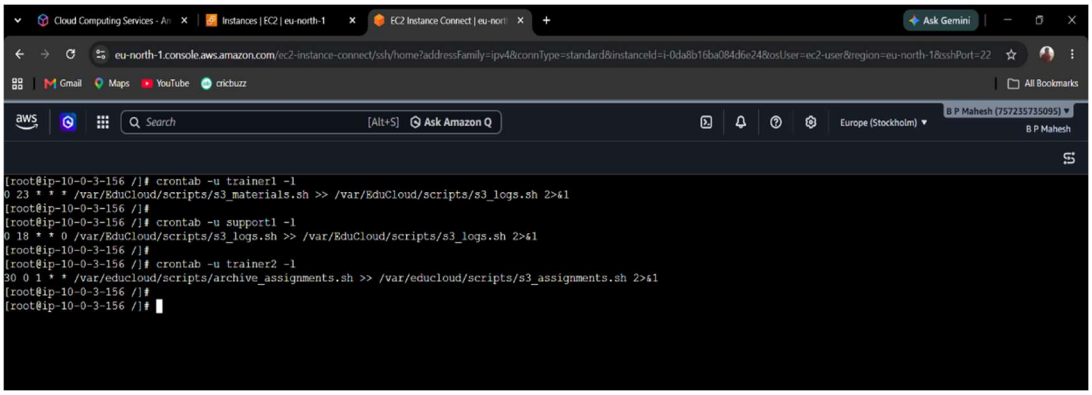
File Handling & Archival

- Trainers archive old course materials into .tar.gz weekly.
- Students submit assignments only as .tar.gz files.



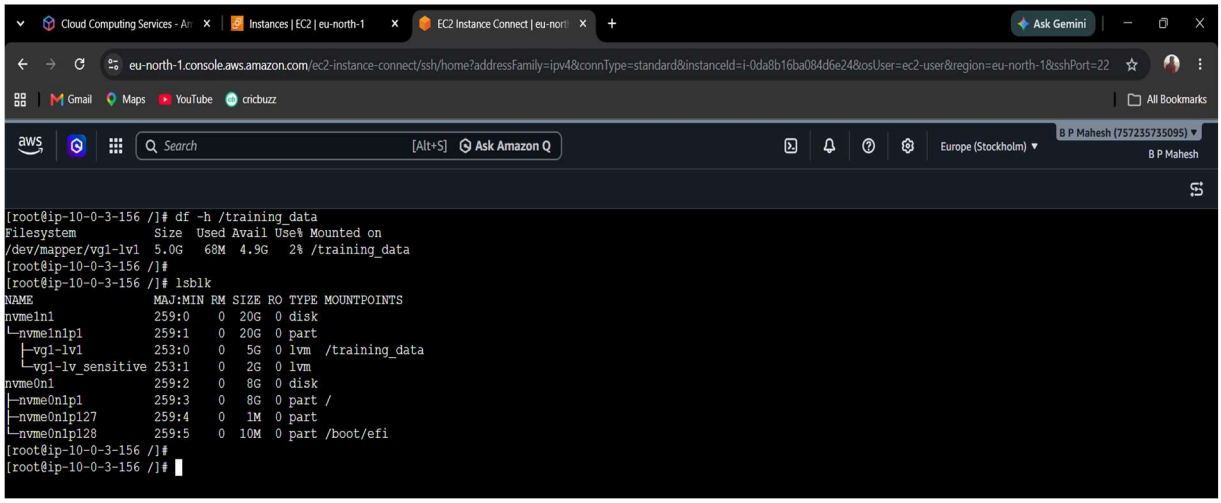
Cron Jobs

• trainer1: Back up /materials daily at 11:00 PM → upload to S3 bucket.
• support1: Upload logs from /logs to S3 every Sunday 6:00 PM.
• trainer2: Rotate assignment submissions into monthly archives on the 1st of every month at 12:30 AM.

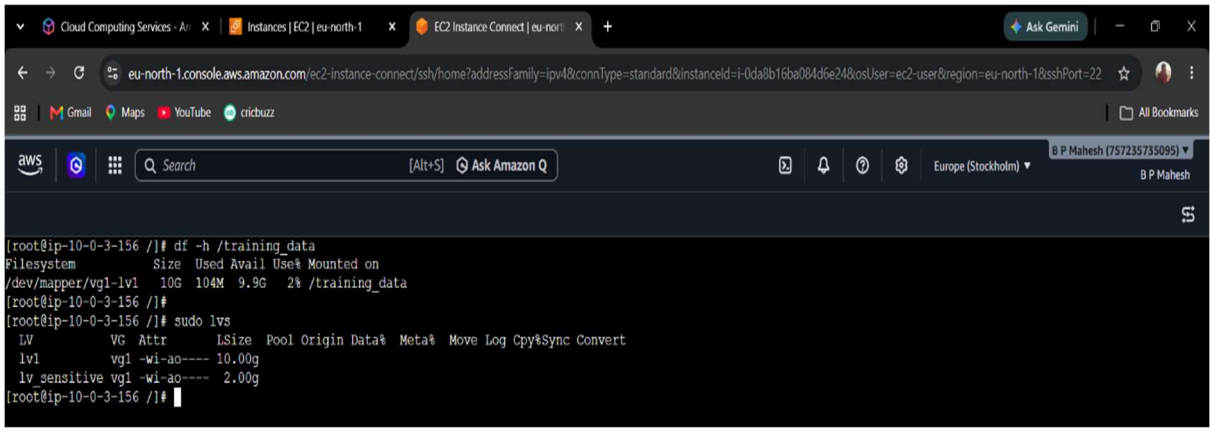


Storage (Partitions & LVM)

- Create /training_data partition of 5GB (students' assignments).
- Secure /sensitive_data partition (2GB) with LUKS encryption (trainer-only).



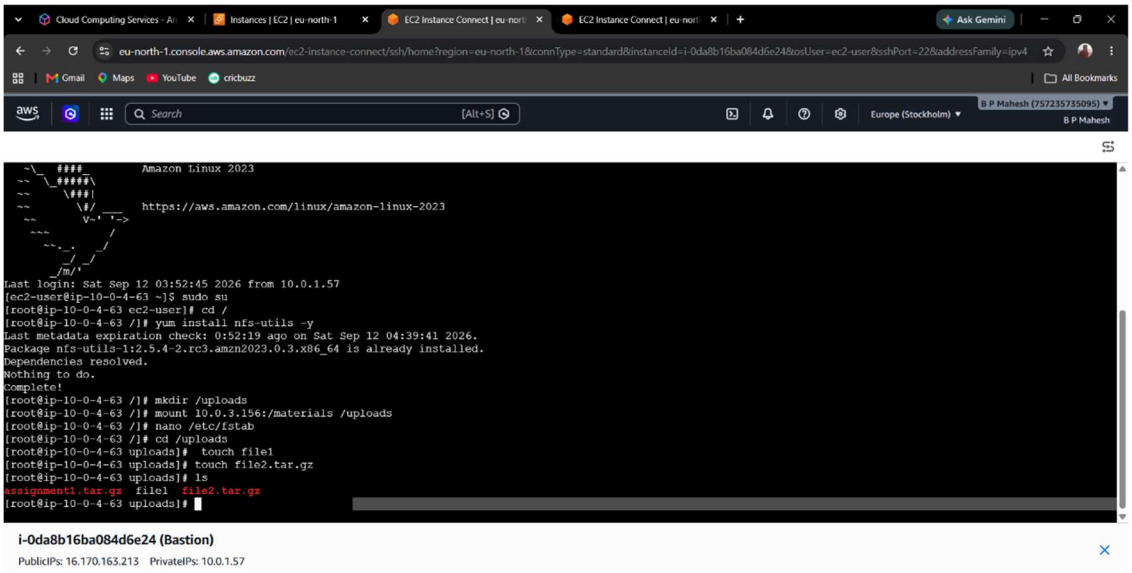
- Expand to 10GB using LVM when assignments exceed 80% usage.
  - sudo lvextend -L 10G /dev/vg1/lv1
  - sudo xfs_growfs /training_data
  - df -h /training_data



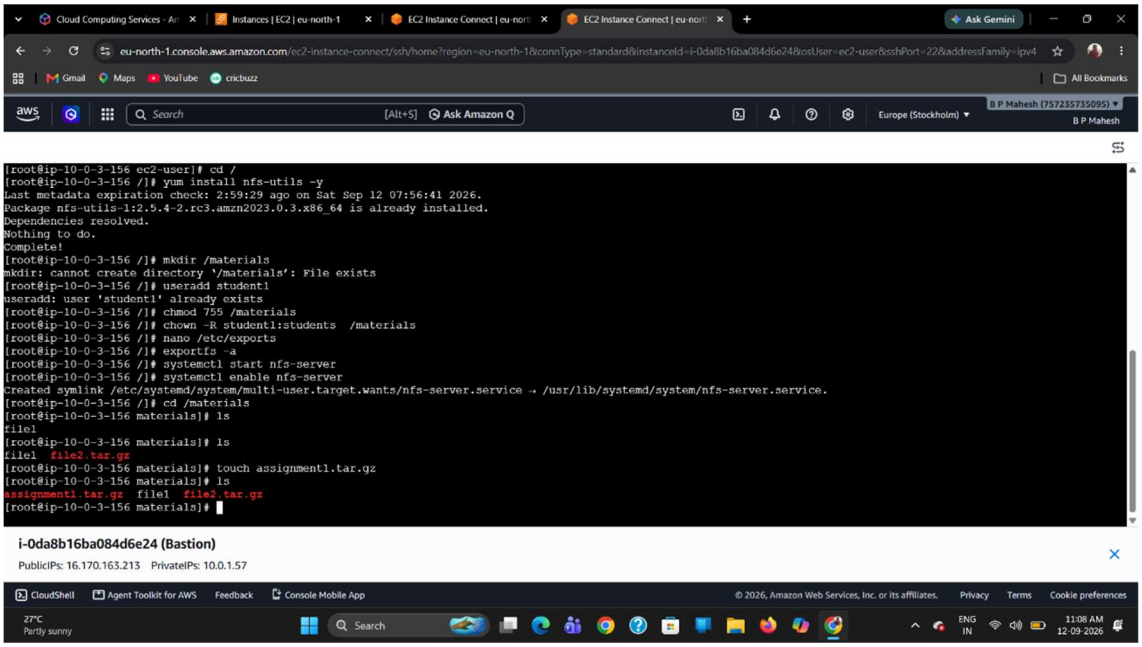
File Sharing

• NFS: /materials → Trainers (read/write), Students (read-only).

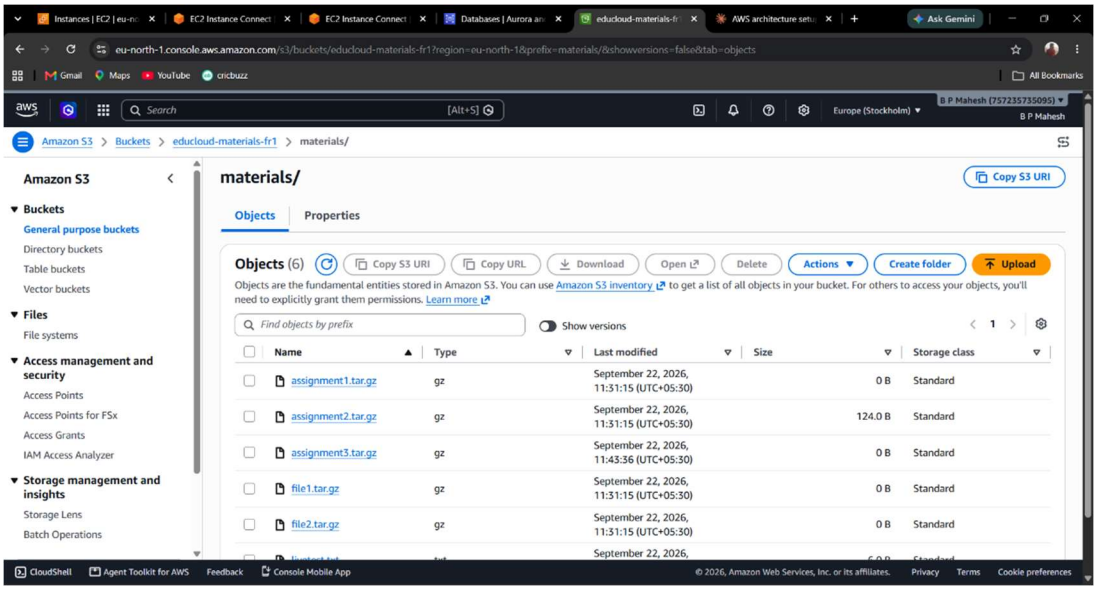
Client:



Server:



Materials Stored in S3-bucket

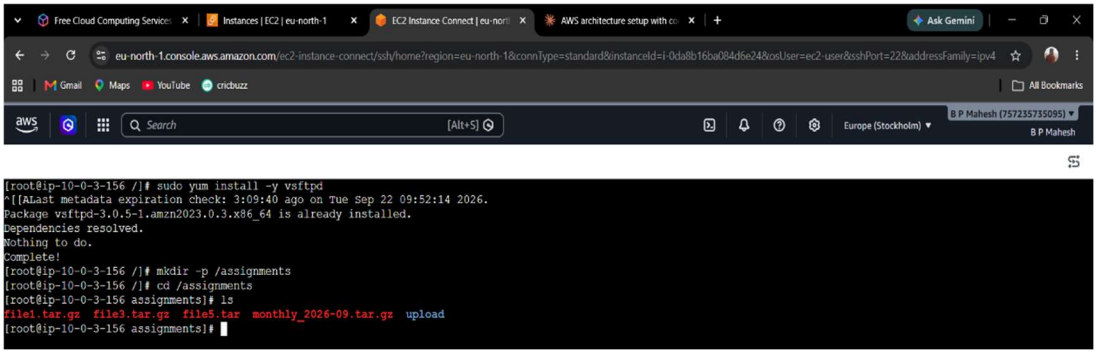


• FTP: Students upload assignments via FTP.

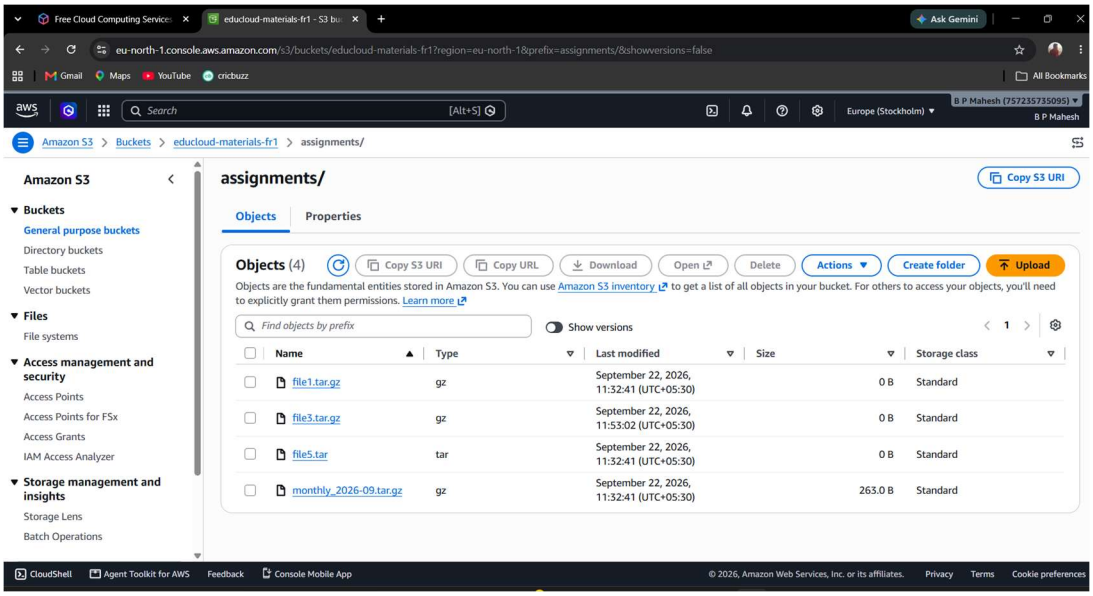
FTP Client:



FTP Server:

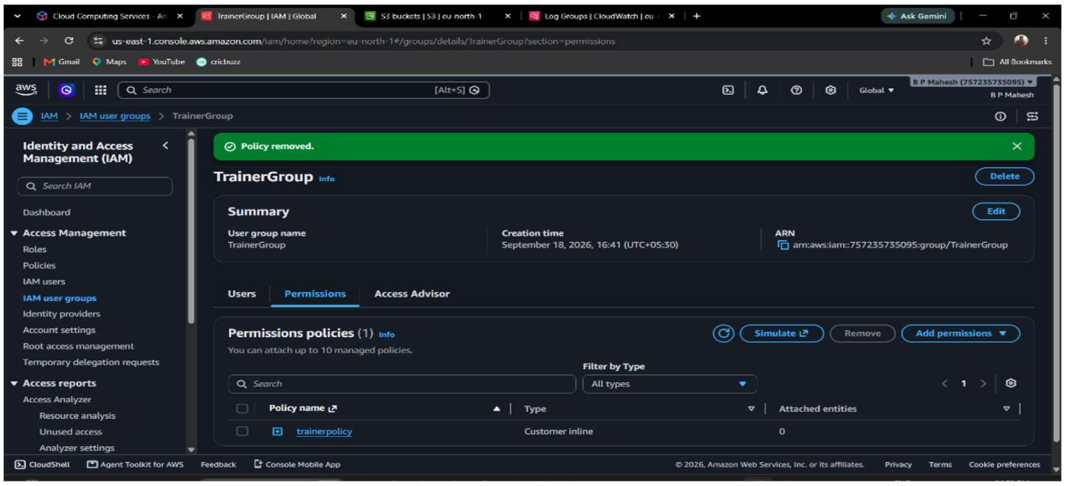


Assignments Stored in S3 Bucket

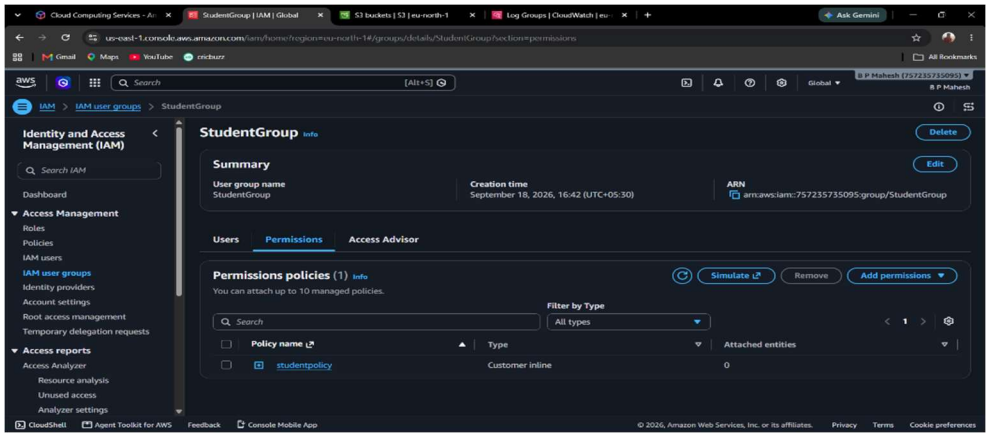


IAM Users:

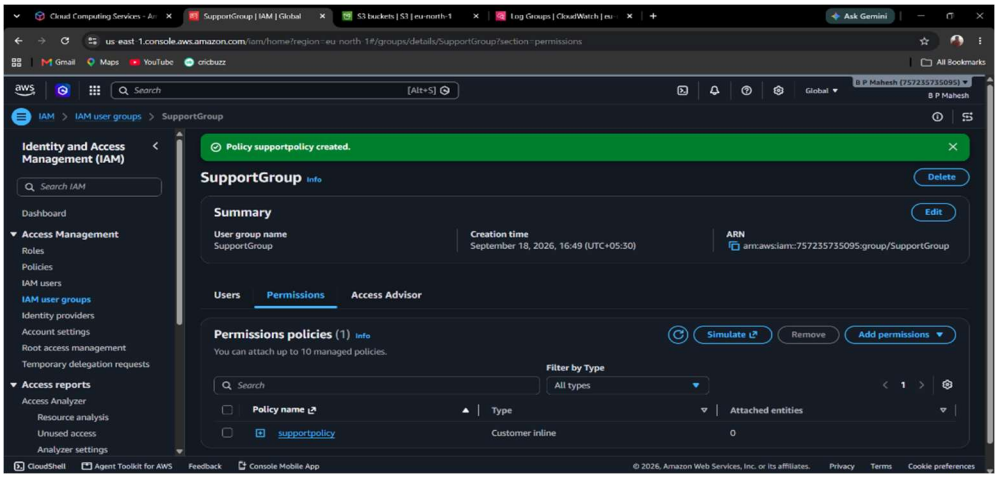
Create groups: TrainerGroup → Full access to S3 bucket educloud-materials.



StudentGroup → Read-only to bucket

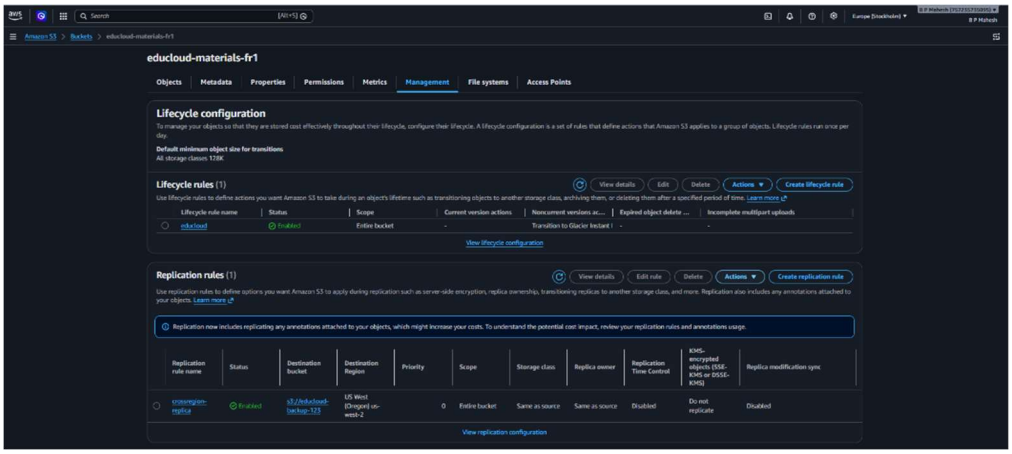


SupportGroup → CloudWatch logs + backup bucket access.

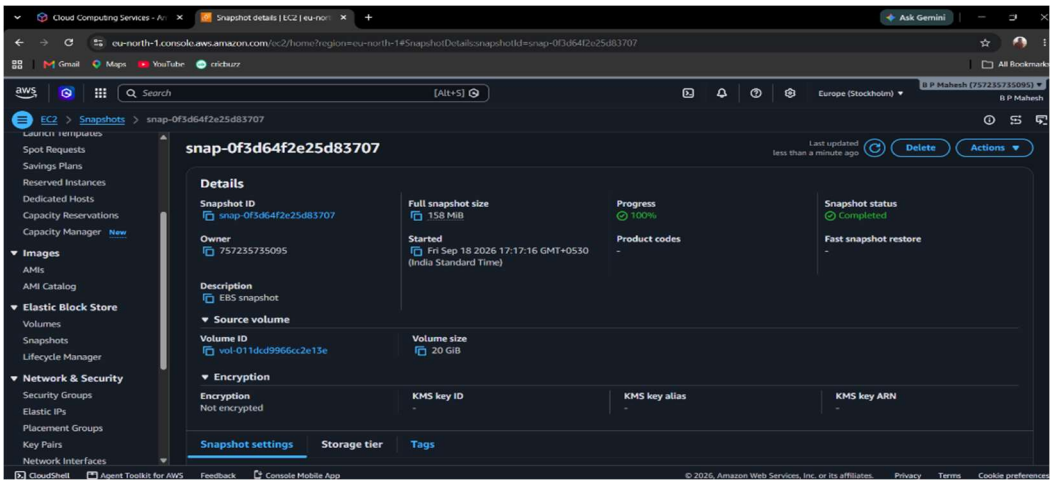


S3 (Materials Storage)

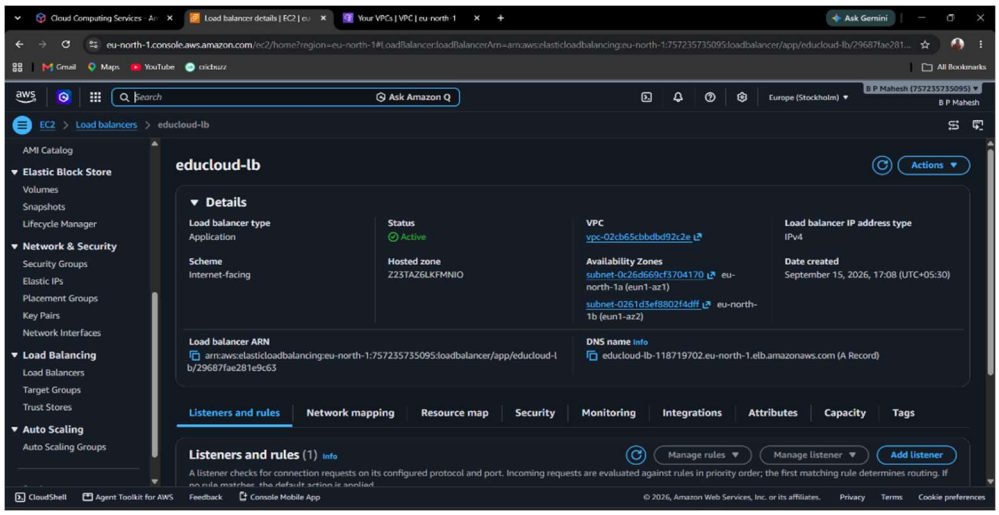
Bucket: educloud-materials Features: Versioning enabled.
Lifecycle policy: Move files older than 90 days → Glacier.
Cross-region replication enabled to backup bucket in us-west-2.



- Attach EBS (20GB) for persistent storage → snapshot weekly.

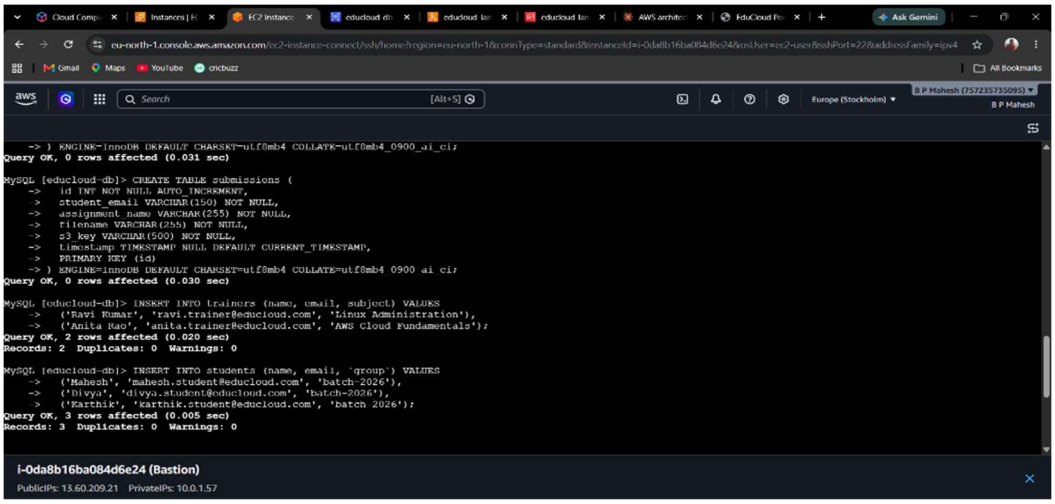


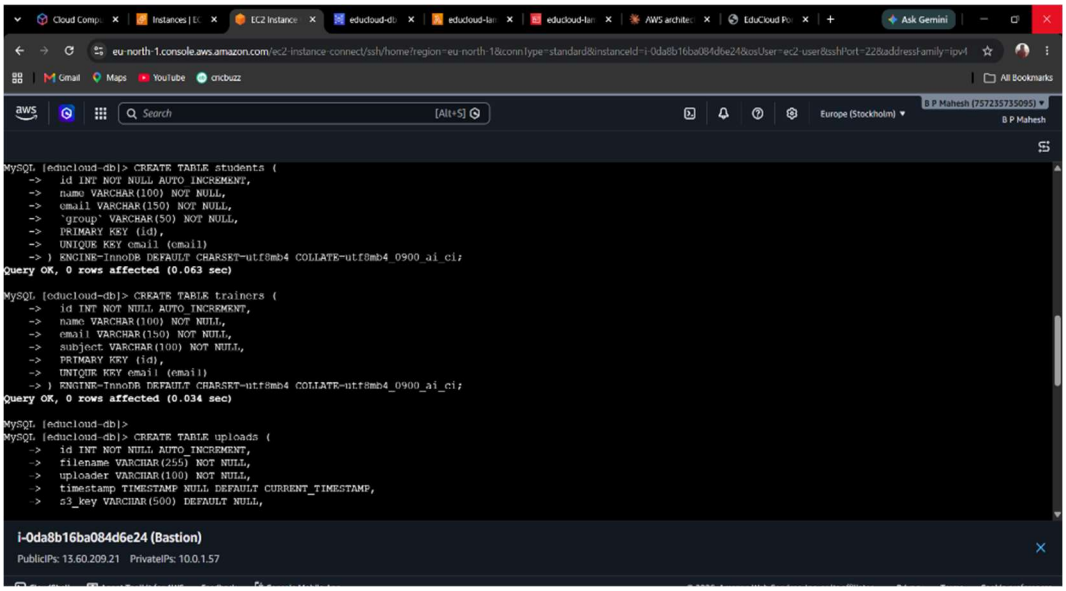
- Place behind an Application Load Balancer (ALB).

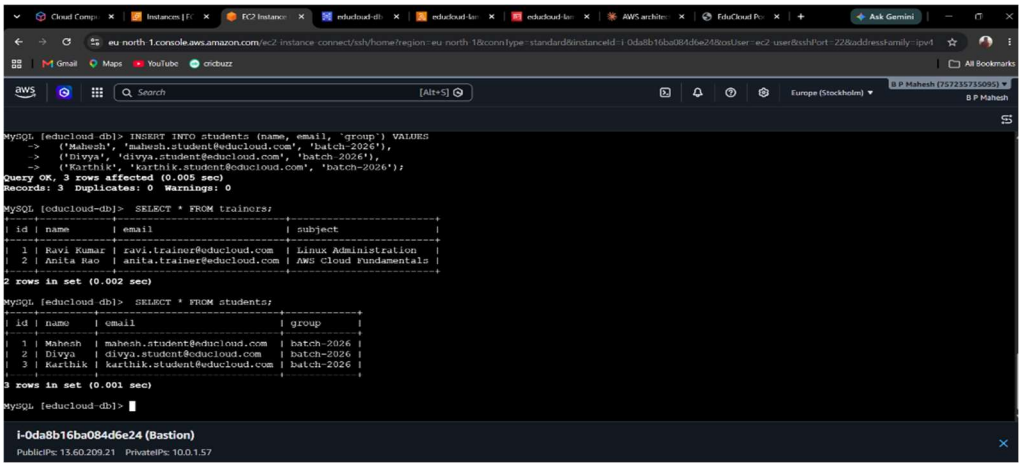


RDS (MySQL)

- Database: educloud_db.
- Tables: students (id, name, email, group)
  trainers (id, name, email, subject)
  uploads (id, filename, uploader, timestamp)
- Trainers can insert data (new students).
- Students can only view their own records.

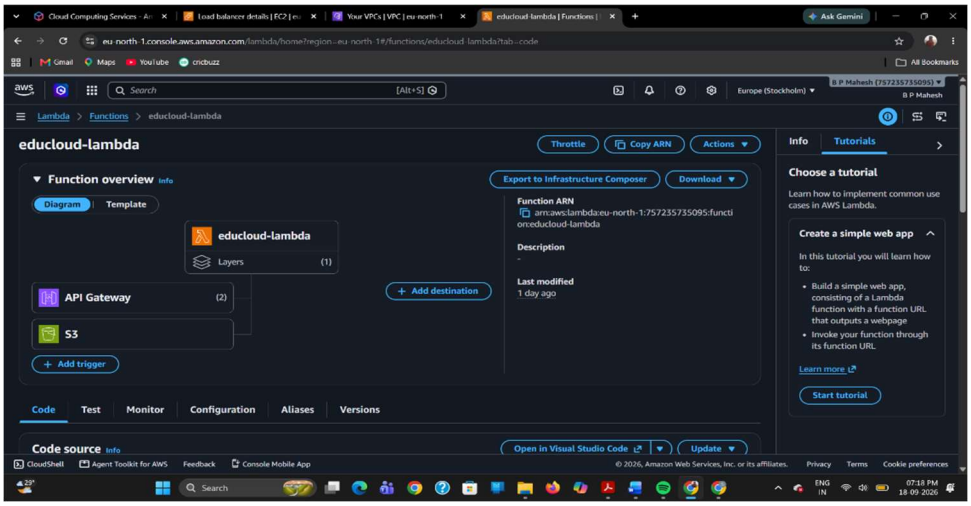




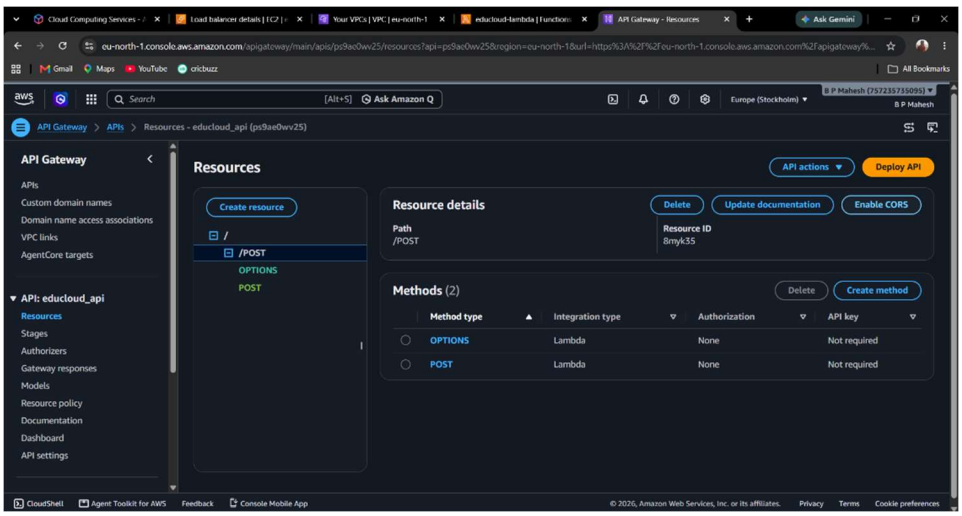


Lambda + API Gateway

• Lambda function: Triggered when a trainer uploads a new file to S3. Sends SNS email notification to all students.

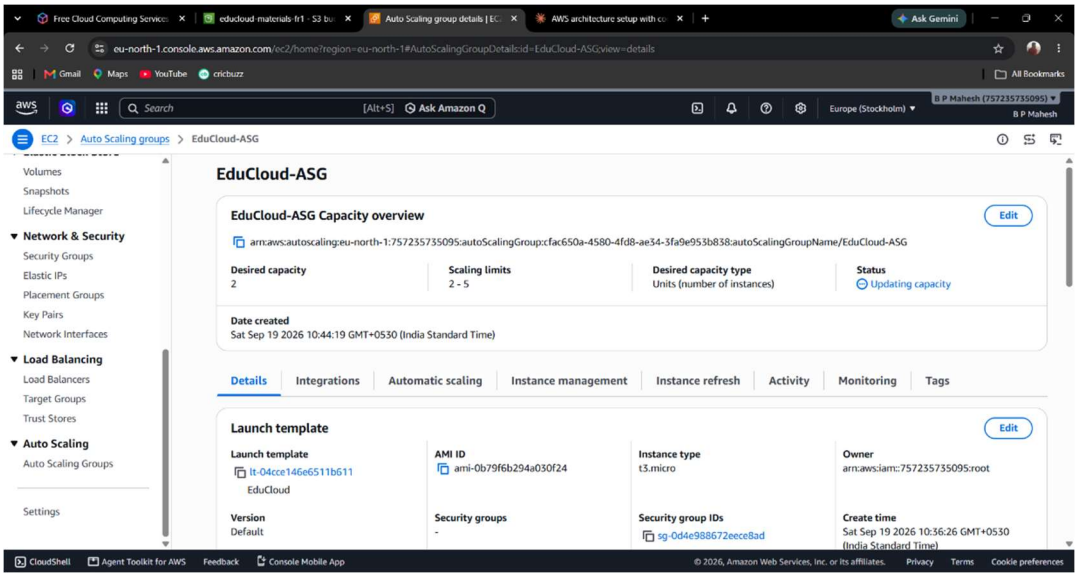


• API Gateway: Expose simple API /submitAssignment. Students hit endpoint → upload metadata into RDS.

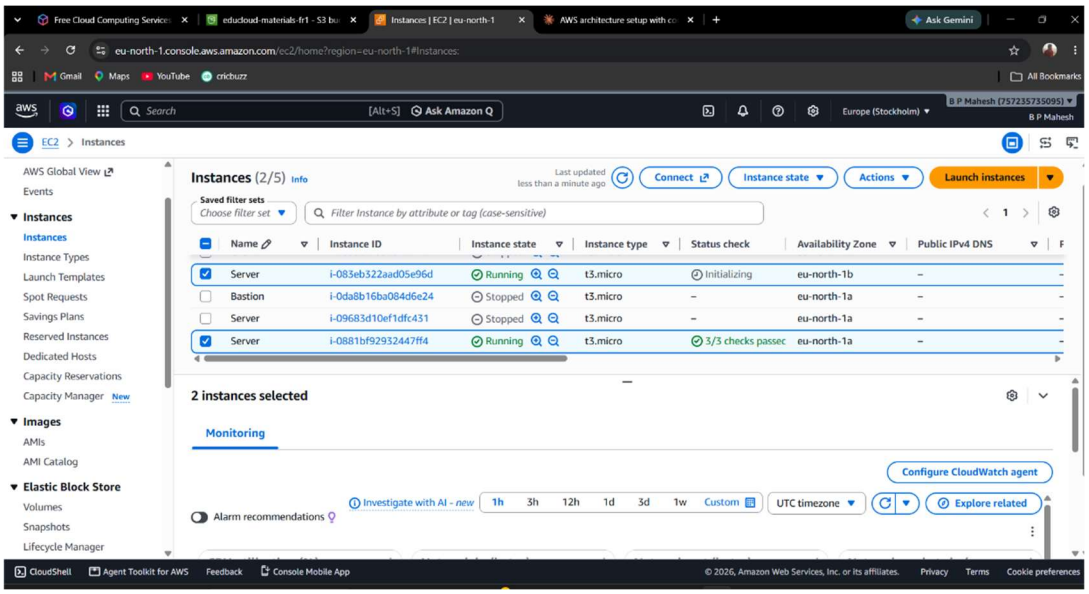


AutoScaling Group

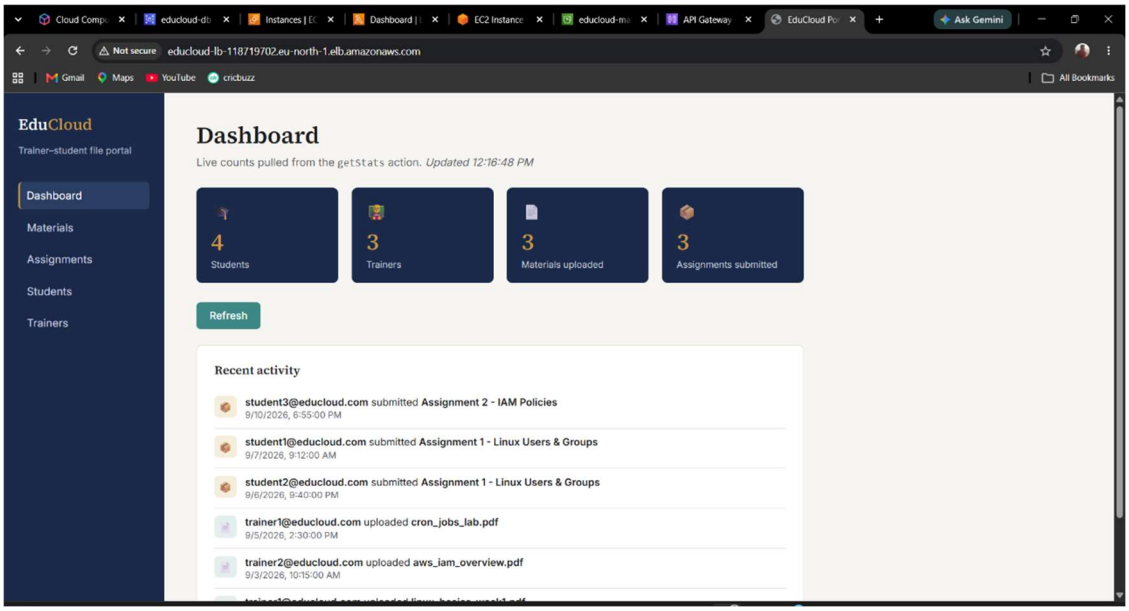
- Minimum: 2 instances.
- Maximum: 5 instances.
- Scale out if CPU > 60% for 5 mins

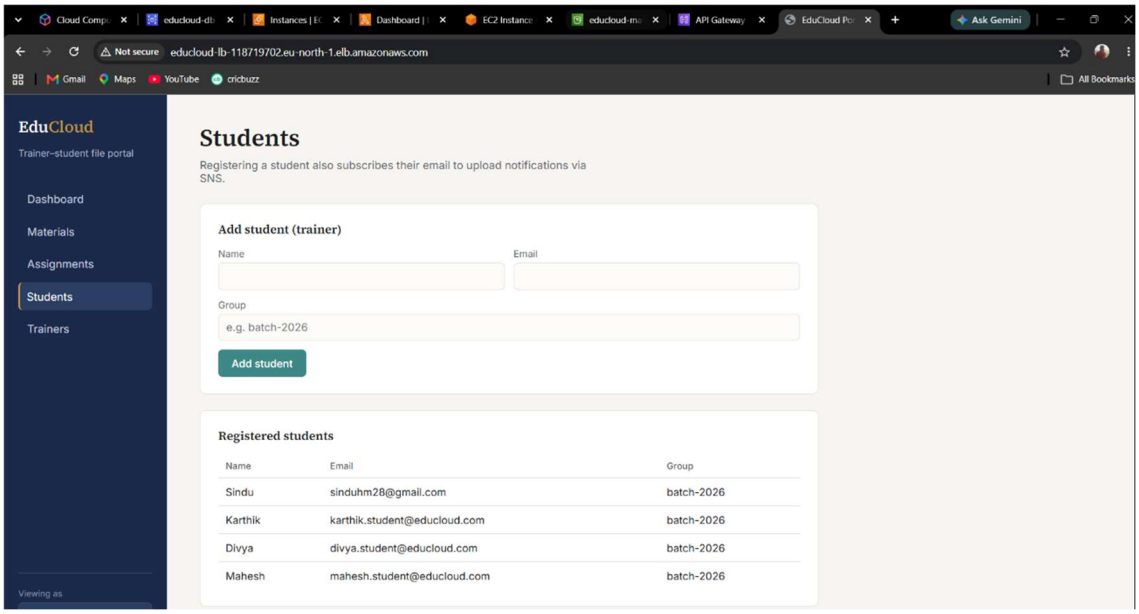


Instance launched Using AutoScaling :



WebPage:





Working DNS EndPoint:

http:// educloud-lb-118719702.eu-north-1.elb.amazonaws.com

Results and Outcomes

The EduCloud project successfully demonstrates the design and implementation of a secure, scalable, and highly available cloud-based educational platform using AWS services. The integration of networking, compute, storage, database, security, content delivery, and automation services resulted in the following outcomes:

Results

- High availability: Application traffic is distributed through the Application Load Balancer (ALB) across multiple EC2 instances, reducing dependency on a single server.
- Automatic scalability: The Auto Scaling Group (ASG) dynamically adjusts EC2 capacity based on workload, helping maintain application performance during changing traffic conditions.
- Secure data storage: Amazon S3 provides centralized and durable storage for study materials, assignment submissions, and backup files.
- Controlled resource access: AWS IAM enables role-based permissions for students, trainers, and support users, following the principle of least privilege.
- Improved content delivery: Amazon CloudFront enables faster delivery of static web content through geographically distributed edge locations.
- Secure networking: The VPC architecture separates public-facing and backend resources using public and private subnets, route tables, Internet Gateway, NAT Gateway, and security groups.
- Database management: Amazon RDS provides managed storage for structured application data and metadata, reducing the need for manual database administration.
- Automation: AWS Lambda and scheduled automation reduce manual intervention by supporting automated processing, notifications, backups, and cloud workflows.
- Backup and recovery: Automated backup and archival mechanisms improve data protection and support recovery in case of accidental deletion or system failures.
- Cost optimization: Managed and scalable AWS services allow resources to be provisioned according to workload requirements rather than maintaining fixed infrastructure.

Project Outcomes

The project provided practical experience in designing and deploying a real-world cloud infrastructure rather than developing only a standalone application. Through EduCloud, the following learning and implementation outcomes were achieved:

1. Practical AWS knowledge was gained through the implementation of EC2, S3, RDS, Lambda, CloudFront, ALB, ASG, IAM, and VPC.
2. Cloud networking concepts such as CIDR addressing, subnetting, routing, Internet Gateway, NAT Gateway, and security groups were implemented practically.

3. High-availability architecture was designed using load balancing, multiple instances, and Auto Scaling.
4. Cloud security practices were applied through IAM roles, controlled permissions, private networking, and security-group rules.
5. Storage and backup management was implemented using S3, lifecycle policies, archival mechanisms, and automated backups.
6. Event-driven and scheduled automation was demonstrated using AWS Lambda and Linux-based automation.
7. Scalable application deployment was achieved by separating frontend, application, database, and storage responsibilities across appropriate AWS services.
8. The project strengthened understanding of cloud architecture, DevOps practices, system administration, monitoring, troubleshooting, and resource management.

Overall Outcome

EduCloud successfully demonstrates how multiple AWS services can be integrated to build a secure, scalable, highly available, and automated educational cloud platform. The project provides a practical understanding of how cloud infrastructure can address the limitations of traditional educational systems while maintaining controlled access, reliable data storage, efficient content delivery, and automated operations.